

Build the beam.

INNOVA PHYSICS UPV · UNIVERSITAT POLITÈCNICA DE VALÈNCIA
PARTNERS FOR AN OPEN-HARDWARE ELECTRON ACCELERATOR

Particle accelerators, open to **everyone**.

Accelerators treat cancer, make chips and probe new materials, yet building one is still the work of national laboratories. ALOHA is a tabletop electron linear accelerator that we design, simulate and publish as open hardware, so any university can build its own.

30 kV

electron source, the first stage

SIMULATED

1 MeV

the machine, designed this season

TARGET

Open

hardware under CERN-OHL-S, KiCad-native

LICENCE

FIG. 01 • ALOHA-0 ELECTRON SOURCE • COLOUR IS KINETIC ENERGY, 0 TO 30 KEV • COMSOL, SIMULATED





TEAM MEMBERS AT THE ACCELECR POSTER SESSION • CERN • JULY 2026

Students who build, from UPV to CERN.

Telecommunications and physics students learn by building a real machine: simulation, high voltage, vacuum, RF and control. Every season ends in front of the accelerator community.



30+

students in four departments OCT 2026

CERN

acceIECR 2026 • PRESENTED
poster session

IPAC'27

student poster at the 2027 • TARGET
main accelerator
conference INNOVA PHYSICS UPV

Join as a partner.

Software, equipment, services or funding. Every contribution is credited where it is used: in our public documentation, on our posters, on the machine.

MINIMUM PER SEASON	Neutrino 100 €	Quark 500 €	Photon 3,000 €	Higgs 10,000 €
Name on our website and public docs	●	●	●	●
Credit in the post about your support	●	●	●	●
Impact letter at the end of the season	●	●	●	●
Name or logo on the team shirt	●	●	●	●
Logo on the website, posts and roll-up	○	●	●	●
Closing slide of our talks	○	●	●	●
Logo on posters and the main roll-up	○	○	●	●
Formal polo and a plate on the machine	○	○	●	●
A named subsystem of the machine	○	○	○	●
A talk at our design challenge	○	○	○	●
Recruiting talk at ETSIT and our CV book	○	○	○	●

Higgs is also by invitation. Licences, loans and donated equipment count: we place them by what they cost you.



WHAT WE NEED THIS SEASON

Simulation licences

COMSOL modules, CST, Ansys, MATLAB

RF instruments

a vector network analyser to test the accelerating cavity

High voltage and vacuum

a 30 kV supply and a vacuum pump cart

Print and apparel

posters, roll-ups and shirts for 30+ members

Let's talk.

Twenty minutes is enough to show you the machine and find where you fit.

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<https://innova-physics-upv.github.io/>

Generación Espontánea · Universitat Politècnica de València